AN INTEGRATED ACADEMIC PLANNING AND COURSE RATING WEB PLATFORM FOR EGCI STUDENTS

MISS TRINNAYA DAMRONGPATHARAWAT
MR. KYAW ZIN THANT
MR.NATHAN TANABORIBOON

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
MAHIDOL UNIVERSITY

2025

AN INTEGRATED ACADEMIC PLANNING AND COURSE RATING WEB PLATFORM FOR EGCI STUDENTS

MISS TRINNAYA DAMRONGPATHARAWAT
MR. KYAW ZIN THANT
MR.NATHAN TANABORIBOON

A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE
BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
MAHIDOL UNIVERSITY
2025

Computer Engineering Project
entitled

AN INTEGRATED ACADEMIC PLANNING AND COURSE RATING WEB PLATFORM FOR EGCI STUDENTS

MISS TRINNAYA DAMRONG-PATHARAWAT
Researcher

MR. KYAW ZIN THANT
Researcher

MR.NATHAN TANABORIBOON
Researcher

Dr. LALITA NARUPIYAKUL, Ph.D. (Computer Science),
Ph.D. (Computer and Electrical Engineering)
Advisor

Thesis
entitled

AN INTEGRATED ACADEMIC PLANNING AND COURSE RATING WEB PLATFORM FOR EGCI STUDENTS

was submitted to the Faculty of Engineering & International College,
Mahidiol University
for the degree of Bachelor of Engineering (Computer Engineering)
on
July 15, 2024

_____
Dr. LALITA NARUPIYAKUL, Ph.D. (Computer Science),
Ph.D. (Computer and Electrical Engineering)
Chair Committee

_____
Committee 1, Ph.D. (Computer Engineering)
Committee

_____
Committee 2, Ph.D. (Computer Engineering)
Comittee

BIOGRAPHY

NAME	MISS TRINNAYA DAMRONGPATHARAWAT
DATE OF BIRTH	26 December 2004
BIRTHPLACE	Nakhon Ratchasima, Thailand
EDUCATION BACKGROUND	Mahidol International College
MOBILE PHONE	098 454 9415
E-MAIL	trinnaya.dam@student.mahidol.edu

NAME	MR. KYAW ZIN THANT
DATE OF BIRTH	11 August 2003
BIRTHPLACE	Yangon, Myanmar
EDUCATION BACKGROUND	Mahidol International College
MOBILE PHONE	082 582 5476
E-MAIL	kyawzin.tha@student.mahidol.edu

BIOGRAPHY

NAME	MR.NATHAN TANABORIBOON
DATE OF BIRTH	17 September 2006
BIRTHPLACE	Bangkok, Thailand
EDUCATION BACKGROUND	Mahidol International College
MOBILE PHONE	093 020 7448
E-MAIL	nathan.tan@student.mahidol.edu

ACKNOWLEDGEMENT

First and foremost, we would like to express our special thanks and sincere of gratitude to our advisor and co-advisors, Asst. Prof. Lalita Narupiyakul, Asst. Prof. YYY YYYY, Asst. Prof. ZZZ ZZZZ, who insight, advise and knowledge us a lot in finalizing this project within the limited time frame.

Beside our advisors, we would like to thank to the committees, Asst. Prof. XXX XXXX, Asst. Prof. YYY YYYY, and Asst. Prof. ZZZ ZZZZ, for their insightful comment and encouragement. Also, the questions which encourage us to rethink and research more about some factors which we missed. Therefore, this project would not be completed without comments, questions, and support from our committees.

Finally, we would like to special thanks to our university, Mahidol University International College, and Faculty of Engineering, Mahidol University, which provided us a lot of resources to study, financial means for researching this project.

MISS TRINNAYA DAMRONGPATHARAWAT
MR. KYAW ZIN THANT
MR.NATHAN TANABORIBOON

ชื่อโครงงาน	ชื่อวิทยานิพนธ์
ผู้วิจัย	นายชื่อผู้แต่ง นามสกุลผู้แต่ง ICCI
	นางสาว ชื่อผู้แต่ง นามสกุลผู้แต่ง ICCI
	นายชื่อผู้แต่ง นามสกุลผู้แต่ง ICCI
	ปริญญา วิศวกรรมศาสตรบัณฑิต สาขาวิชาวิศวกรรมคอมพิวเตอร์
อาจารย์ที่ปรึกษา	ดร. ชื่อที่ปรึกษา นามสกุลที่ปรึกษา
สำเร็จการศึกษา	15 กรกฎาคม 2569

บทคัดย่อ

บทคัดย่อภาษาไทย (not required for your project proposal, but it is mandatory for the black book)

คำสำคัญ : ลาเท็ก / วิทยานิพนธ์ (~5 คำ)

16 หน้า

AN INTEGRATED ACADEMIC PLANNING AND COURSE RATING WEB PLATFORM FOR EGCI STUDENTS

STUDENTS	MISS TRINNAYA DAMRONGPATHARAWAT ICCI
	MR. KYAW ZIN THANT ICCI
	MR. NATHAN TANABORIBOON ICCI
DEGREE	Bachelor of Engineering (Computer Engineering)
PROJECT ADVISOR	Dr. LALITA NARUPIYAKUL, Ph.D. (Computer Science),
	Ph.D. (Computer and Electrical Engineering)
DATE OF GRADUATION	July 15, 2024

ABSTRACT

AN INTEGRATED ACADEMIC PLANNING AND COURSE RATING WEB PLATFORM FOR EGCI STUDENTS

An abstract (250-400 words) should provide a concise summary of your entire thesis. It should report significant elements of your thesis including background or introduction in brief, objectives, statistical data, key findings, and conclusion. Mathematical formulas, diagrams, and other illustrativematerials are not recommended for inclusion. A strong abstract should be self-contained; without abbreviations, footnotes, references. Outside readers typically view the abstract before deciding to read the thesis, so it should be well written, logical, and a complete reflection on you work. The abstract is typically written last after finishing chapter 4 and 5.

KEYWORDS : LaTeX / Thesis (~5 words)

16 Pages

CONTENTS

	Page
BIOGRAPHY	**i**
ACKNOWLEDGEMENT	**iii**
ABSTRACT (THAI)	**iv**
ABSTRACT (ENGLISH)	**v**
CONTENTS	**vi**
LIST OF TABLES	**viii**
LIST OF FIGURES	**ix**
CHAPTER 1 INTRODUCTION	**1**
1.1 Background	1
1.2 Objective	2
1.3 Scope	3
1.4 Expected Results	4
1.5 Timeline	4
CHAPTER 2 LITERATURE REVIEW	**5**
2.1 Design web service academic information system based multiplatform [9]	5
2.2 Design and Implementation of a Digital Notice Board and Forum Discussion System [11]	6
2.3 Course Selling Website Using MERN Stack [12]	6
2.4 CourseViewer a prototype for visualizing undergraduate courses and student transcripts [3]	7
2.5 The curriculum prerequisite network: a tool for visualizing and analyzing academic curricula [2]	8
2.6 A hyper-heuristic algorithm based on genetic and greedy strategy for university course scheduling problem [13]	8
2.7 A Review of Strategies for Designing, Administering, and Using Student Ratings of Instruction [7]	9

CHAPTER 3 METHODOLOGY **11**
3.1 System Design 11
3.2 Data 12
3.3 Methods 12
3.4 Hyperparameter Configurations 12
3.5 Performace Evaluation 12
CHAPTER 4 RESULT **13**
4.1 Results 13
4.2 Discussions 13
CHAPTER 5 Conclusion **14**
5.1 Conclusion 14
5.2 Obstacles 14
5.3 Future Work 14
REFERENCES **15**

LIST OF TABLES

Table **Page**

1.1 Project Timeline 4

LIST OF FIGURES

Figure **Page**

3.1 Paragraph arrangement (example) 11

CHAPTER 1
INTRODUCTION

1.1 Background

Academic planning is important especially for engineering students who must follow a structured curriculum with multiple prerequisites and limited courses offered each semester. Evaluating course difficulty, reviewing the course curriculum and coordinating schedules with peers for group study are also the struggles that most students are facing in the faculty. Finding an available time slot between friends is also a time-consuming struggle as students need to look up each other's schedule constantly.

At the EGCI program, students currently lack a centralized platform that allows them to rate the course, find available time through class-schedule screenshot and plan their academic paths in a visual and structured way. Course feedback is only shared when students ask questions in LINE group chats and there is no platform where information can be accessed any time. Additionally, comparing schedules with friends to find common free time requires manual checking which is time consuming and error prone when there are multiple students. The process of just using the screenshots of the schedule and finding the available times would be a great solution for meetings and collaborations. Students also face difficulties when planning their academic paths. Most students rely on the student handbook to check course requirements, prerequisites, and credit conditions. However, the handbook only serves as a reference document and does not provide interactive guidance. As a result, many students become confused about which general education courses they still need to take, which electives are required, and how their course selections affect their graduation timeline. It is common for students to enroll in available courses without a clear long-term plan, which may lead to delayed graduation or inefficient course sequencing.

Several systems such as online course platforms and academic planning tools have been developed to address individual aspects of this problem. For example, "RateMyProfessors" website [1], developed to allow students to share their opinions and comments of their professors with colleagues, or the Stellic and DegreesWork [4] [6], academic planning websites, designed to let students stay informed of their academic performance and possible academic career path that they can choose. However, most of these systems focus on single functions and lack certain capability to solve our addressed issues, for example, [1] lack a lot of information about EGCI curriculum structures, Stellic and DegreesWork [4] [6] use text-based academic planning, which [10] addressed that graph-based representations of course prerequisites can improve student's understanding of course syllabus. In addition, the issues of the manual schedule coordinations of students can also be solved by the combination of TableExtractNet, by Ngubane and Tapamo [8] a deep learning approach utilizing CornerNet and Faster R-CNN to detect and recognize table structures from document images, which can be implemented and trained to turn the photos/screenshots of class schedules into a digital form, utilizing the "Master Busy Map" logic [5] to further develop an algorithm to discover the common available of many class schedules.

The objective of this project is to develop EGCEye, an integrated academic planning and course rating web platform specifically for EGCI students. The system will include three main functions: a class schedule matcher using image processing techniques, a rate-my-course feature for student reviews and feedback, and a super visualized planner that guides students through their academic progress toward graduation.

1.2 Objective

1. To develop a website for EGCI students to rate courses and leave comments on the courses offered by MUIC and the EGCI faculty, share their class schedules with friends to find common available times, and create an academic plan to track both courses they intend to enroll in and those they have already completed throughout their studies with the EGCI faculty.

1.3 Scope

1. The Optical Character Recognition will be used to read any text labels of date and time from the class schedule screen capture.
2. The coloured blobs will be detected by OpenCV. We will perform color segmentation to extract the class block, which will then be put in Master Busy Map to perform OR logic operation to find common free times among students.
3. If no common available time is found for the entire group, the system will suggest available time slots for smaller subgroups by excluding one student at a time and identifying overlapping free periods among the remaining members
4. Allows EGCI students to rate the any available General Education courses, major required course, and engineering core course offered by MUIC and EGCI faculty, as a score from 0 to 5 and allows users to leave a comment anonymously.
5. List of courses available to rate will be initially created by us but users can also choose to add more choice of courses in case new courses are available.
6. Any available courses will also be separated by professors in case the course has more than one section, which are taught by different professors.
7. The function offers graphical interfaces for users to arrange the plan for their entire academic studies with the EGCI faculty, the computer engineering major required courses and engineering core courses will be in a form of a floating icons which users can drag and place them in a slot for each semester, with multiple semesters available. Each course that has prerequisite courses will have a line connecting to the prerequisite courses. There would also be an additional "suggestion" line, which connects between a course and other available courses which are suggested by the faculty to study first although the course is not connected by the prerequisite line. With the general education courses, I-design courses, and major electives courses, the website will allow users to specify what course they would take to complete the required credit of that category.
8. The courses that are not yet placed in the plan will also display a marker to suggest the users of their choice of course enrollment, for example, a second year course that has never been enrolled by the third year student will show a marker and a suggestion to the student that they should consider taking it as soon as possible.

1.4 Expected Results

1. The proposed system is expected to achieve a user base of 50 or more.
2. The Rate-My-Course reaches the number of activities: 15 for adding star-rating, and 5 for leaving comments.
3. The Class Schedule Matcher targets a number of 15-20 screenshots uploaded to find common free times, with all being accurately matched.
4. The Super Planner aims to facilitate at least 5-10 unique student plans. 2-3 trimesters (slots) or more mapped out are required to be counted as one plan to preserve its long-term plan characteristic.
5. The project aims for an average User Experience (UX) rating of 4.0-4.5 out of 5 stars based on quality feedback collected from a random group of users.

1.5 Timeline

Table 1.1 Project Timeline

Plan	**Timeline 2026**										
	Feb	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
1: Literature Review											
2: System Arch. Design											
3: UI/UX Deign											
4: Rate My Course											
5: Schedule Matcher											
6: Super Planner											
7: Testing											
8: Documentation											
9: Final Presentation											

CHAPTER 2
LITERATURE REVIEW

This chapter discusses the published academic papers used to support our project. There are two main parts: The first part discussed the technology stack, website development framework, and website development planning that has been implemented and met with success. Such technology stack and planning would then be used as a guideline to our project development. The second part explores any methods or algorithms that could be adapted and used for the project's main functions development, which would likely be implemented in each function to help fasten the project development.

2.1 Design web service academic information system based multiplatform [9]

This paper proposed a system design that uses Web Services to make academic information available on both web and mobile platforms.

The core of the design is a REST API that uses JSON to exchange data between the server and different types of devices. This approach was chosen because it allows a single database to support many different user interfaces while keeping all the information synchronized in real-time.

The study found that using this web service architecture makes the system much easier to maintain and scale up because the "business logic" on the server is kept separate from how the app looks on the screen.

For future work, the authors suggested adding stronger security protocols to the API and making the database faster so it can handle more students using the app at the same time.

2.2 Design and Implementation of a Digital Notice Board and Forum Discussion System [11]

This paper presents an educational website, equipped with two functions, digital notice board and discussion forum, which are developed to fulfill two goals that aim to improve university students' quality of life. The first goal is to create a centralized space for information dissemination, to ensure that any relevant parties can stay informed of any news or announcement. Second goal is to create an easy-to-access educational digital discussion platform in order to foster communication and discussions among students university staff.

Technically, the website was built using MERN stack (MongoDB, Express.js, React.js, and Node.js), which is a website development framework that is famous for its quick development time, ease of use, and its compatibility with JavaScript language, with the framework being complete on its own, covering front end, back end, API, and database, all within one framework.

The research concludes that transitioning to a digital framework significantly improves user engagement and reduces the risk of miscommunication. For future development, the author suggests further optimizing the user experience and exploring additional features.

As the paper provides information of how the website was planned and developed, including the technology stack and software development model used during the planning phase, such information will be used as a suggestion to our project planning, as the paper has proven itself to be successful, we hope that some part of the paper's strategy in website development could be implemented with our project as well.

2.3 Course Selling Website Using MERN Stack [12]

This paper presents an educational and commercial website that lets university teacher to create, manage, and sell their courses online, intended to be a platform that improves the education quality by narrowing the gap between student's demand for educational courses and teacher's courses supply under the platform equipped with firm security and secure user's privacy protection.

With the website development heavily implements the MERN stack (MongoDB, Express.js, React.js, Node.js) which is a website development framework that is known for its quick development time, ease of use, and its high compatibility with JavaScript language. Additionally, the website also implements the REST API to obtain website's fast response time, and the capability to handle a large number of users and website's course data without being slowed down.

The study found that this architecture makes the platform very flexible, allowing for features like real-time chat and secure payments to be added easily. By keeping the "business side" of the website separate from the "user side," the system is easier to update as more students join.

For future work, the author suggests adding mobile options for the website, AI assistance to help answer student's questions, and digital certificate to serve as a proof when students finish their courses.

With the paper's outlined website architecture and REST API implementation and its prove of effectiveness, such approach to educational website development can be expected to be implemented with our project.

2.4 CourseViewer a prototype for visualizing undergraduate courses and student transcripts [3]

This paper introduces CourseViewer, a prototype that transforms traditional, text-heavy transcripts into interactive visual maps to help students plan their academic paths. This approach is highly relevant for an academic planner because it replaces manual checking with a dynamic system that shows how different courses are connected.

The most useful feature for a modern planner is the graph-based visualization. By representing courses as "nodes" and prerequisites as directed arrows, the system makes it easy to see which classes "unlock" future subjects. This technical design helps prevent enrollment errors by clearly showing if a student has met the necessary requirements before they try to add a class.

Technically, the tool uses the Java language and the Prefuse visualization toolkit to automatically handle the map layout so the user doesn't have to organize it manually.

It also uses color-coding—such as green for completed and red for required—which provides an immediate visual status of a student's progress toward graduation.

The study found that these visual maps allow students to identify their remaining credits much faster and more accurately than reading a paper transcript. For future work, the authors suggested adding a "what-if" feature to simulate different majors and integrating the tool directly into online registration systems.

2.5 The curriculum prerequisite network: a tool for visualizing and analyzing academic curricula [2]

This paper treats a university curriculum as a "complex system" that can be mapped using network science to reveal the hidden connections between courses. This is directly applicable to the project's goal of automating 4-year academic planning for major and general education courses, which currently must be done manually.

The study uses a Directed Acyclic Graph (DAG) where courses are "nodes" and prerequisites are "links". A key technical detail is the identification of "hubs" (courses that unlock many others) and "bridges" (links between different course groups). Implementing this logic allows the interactive planner to visually flag "bottleneck" courses, helping students avoid choices that could delay their graduation by a year.

Technically, the research utilized NetworkX and Gephi to analyze network density and "betweenness centrality". In the project, these metrics can be used to calculate the "long-term impact" of taking a specific class, providing an automated "roadmap" that ensures all prerequisite constraints are met at a glance.

2.6 A hyper-heuristic algorithm based on genetic and greedy strategy for university course scheduling problem [13]

This study addresses the University Course Scheduling Problem (UCSP) by proposing a novel model that balances resources like classrooms and teachers while satisfying complex constraints.

A hyper-heuristic algorithm (GA_RG_HH) was developed, combining Genetic Algorithms (GA) with greedy strategies to optimize scheduling efficiency.

Performance was evaluated using real-world datasets, measuring the reduction of soft constraint violations and the efficiency of resource allocation.

The experiment indicated that the proposed GA_RG_HH performs better than traditional algorithms like Tabu Search and Particle Swarm Optimization. This is attributed to the algorithm's ability to balance exploration and exploitation through a pre-check strategy and dynamic operator sequences.

Future work includes further optimization of the algorithm for even larger-scale institutional data and the inclusion of more diverse constraint parameters.

2.7 A Review of Strategies for Designing, Administering, and Using Student Ratings of Instruction [7]

This paper reviews the best ways to design and use student feedback (course evaluations) to help teachers improve their classes, specifically in medical and pharmacy programs.

The core of the study suggests that a good evaluation system should be carefully planned to be fair and useful. The authors recommend using standardized online forms with about 10 to 30 questions that are easy for students to understand. This approach was chosen because online tools are faster, save paper, and make it easier for teachers to see their results quickly. To keep students from getting "survey fatigue," the study suggests giving them dedicated time during class to finish the evaluations.

The research found that these evaluations are most effective when they focus on "scholarly teaching"—meaning the questions ask about clear goals, good preparation, and how well the teacher presents the material. The study also notes that while online evaluations might have fewer students responding than paper ones, the actual scores and feedback are usually the same.

For future improvements, the authors suggest that schools should use random sampling to get reliable feedback without asking every student every time, create a clear plan for who gets to see the results and how they will be used to help teachers grow, train mentors to help faculty members understand their feedback and make positive changes to their courses.

NOT SURE WHETHER TO INCLUDE OR NOT

CHAPTER 3
METHODOLOGY

Describe what you did so that others can follow to recreate your work/system/ experiment proposed in this project. You should provide addiquet details about your work, system, algorithm, or design. This includes technical methods used (e.g., machine learning techniques, library, tools), data collection and preparation/pre-Processing, hyperparameter configurations, measurements, evaluation approaches or techniques used to evaluate your work performance. (It is recommended to writing this chapter after completing CH2 Literature Review)

3.1 System Design

You may start with an overview of your system/algorithm design by using the followings:

- Flowchart
- System, Design, or Block Diagram
- Phedu Code
- Others (figures or even tables)

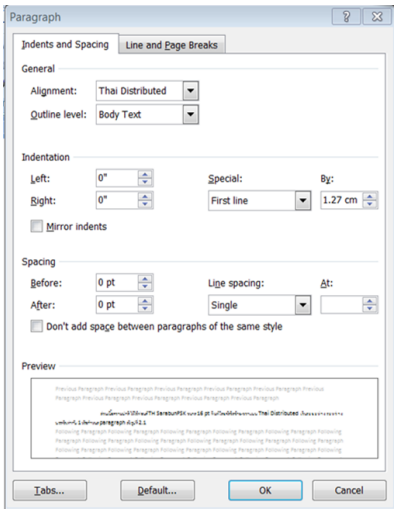

Figure 3.1 Paragraph arrangement (example)

3.2 Data

3.3 Methods

3.4 Hyperparameter Configurations

3.5 Performace Evaluation

CHAPTER 4
RESULT

This chapter reports your experiment results or study report proposed in previous chapter. It typically consists of 1) Results in terms of tables or figures and 2) explanation or discussion of your results, study, or key findings.

4.1 Results

(Experiment/Results in the forms ofrtables or/and figures)

4.2 Discussions

(Interpretation or the meaning of your experiment/results)

CHAPTER 5
CONCLUSION

This chapter summarizes all of your work. It typically starts with the brief explanation of of what you do such as the objective of your work, design, data, experiment results, key findings, interpretation of your experiment and/or result. The obstacles of your work can also be discussed and finally followed by future work.

5.1 Conclusion

...

5.2 Obstacles

...

5.3 Future Work

...

REFERENCES

[1] Rate my professors. https://www.ratemyprofessors.com/. Accessed: 2024-05-20.

[2] P. R. Aldrich. The curriculum prerequisite network: a tool for visualizing and analyzing academic curricula. *arXiv preprint arXiv:1408.5340*, pages 1--10, 08 2014. doi: 10.48550/arXiv.1408.5340. Subjects: Social and Information Networks (cs.SI); Computers and Society (cs.CY).

[3] M. R. de Oliveira, P. J. C. de Mendonça, A. F. G. Leardine, V. P. Gonçalves, B. F. Medina, and C. G. da Silva. Courseviewer -a prototype for visualizing undergraduate courses and student transcripts. In *Software Engineering and Information Systems Laboratory, School of Technology, University of Campinas*, Limeira, Brazil, 2018. Email: celmar@ft.unicamp.br.

[4] S. Inc. Stellic: Degree management and student success platform. https://www.stellic.com/, n.d. Accessed: 2025-02-11.

[5] H. Jiang and B. Mike. Preference-based group scheduling. 3585:990--993, 09 2005. doi: 10.1007/11555261_87.

[6] E. C. L.P. Ellucian degree works. https://www.ellucian.com/solutions/degree-works, n.d. Accessed: 2025-02-11.

[7] M. S. Medina, W. T. Smith, S. Kolluru, E. A. Sheaffer, and M. DiVall. A review of strategies for designing, administering, and using student ratings of instruction. *American Journal of Pharmaceutical Education*, 83(5):7177, 06 2019. doi: 10.5688/ajpe7177. PMCID: PMC6630867; PMID: 31333266.

[8] T. Ngubane and J.-R. Tapamo. Tableextractnet: A model of automatic detection and recognition of table structures from unstructured documents. *Informatics*, 11(4):77, 2024. doi: 10.3390/informatics11040077.

[9] M. L. Pandini, Z. Arifin, and D. M. Khairina. Design web service academic information system based multiplatform. In *2014 The 1st International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)*, pages 297--302, Semarang, Indonesia, 11 2014. doi: 10.1109/ICITACEE.2014.7065760.

[10] M. Trippel and R. Röpke. Developing a graph-based visualization of elective courses to support course selection in higher education, 2025. URL https://dl.gi.de/handle/20.500.12116/47141.

[11] T. Vaidya. Design and implementation of a digital notice board and forum discussion system. *Global International Multidisciplinary Research Journal (GIMRJ)*, XII(I(VIII)):P097, 06 2024. ISSN 2394-8426. doi: 10.69758/GIMRJ2406I8V12P097. Special Issue On Advancements and Innovations in Computer Application.

[12] M. K. Warsi, K. Baghel, C. Gupta, and S. Alphonse. Course selling website using mern stack. In *2025 International Conference on Innovative Trends in Information Technology (ICITIIT)*, pages 1--5, Kottayam, India, 02 2025. doi: 10.1109/ICITIIT64777.2025.11040918.

[13] Z. Zhu, Y. Zhou, and Q. Luo. A hyper-heuristic algorithm based on genetic and greedy strategy for university course scheduling problem. *Applied Soft Computing*, pages 1--26, 2025. doi: 10.1016/j.asoc.2025.114150.